

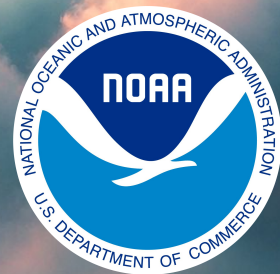
# Makara Passive Acoustics Database & PACM

PAM SI Workshop  
September 24, 2026

Jeffrey D. Walker, PhD

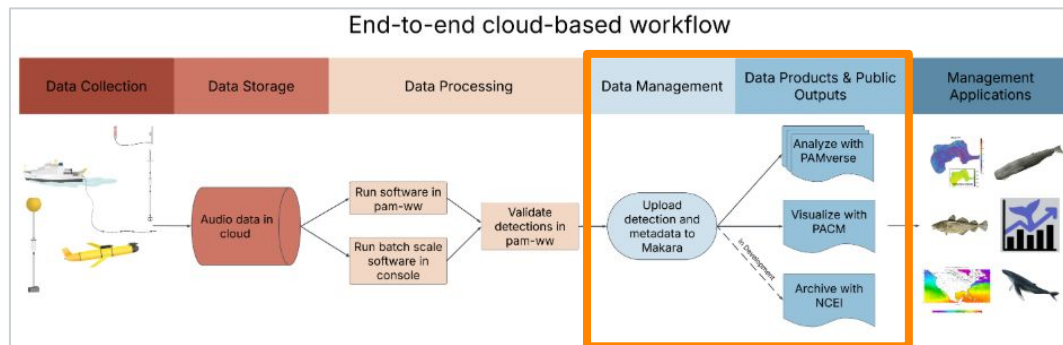
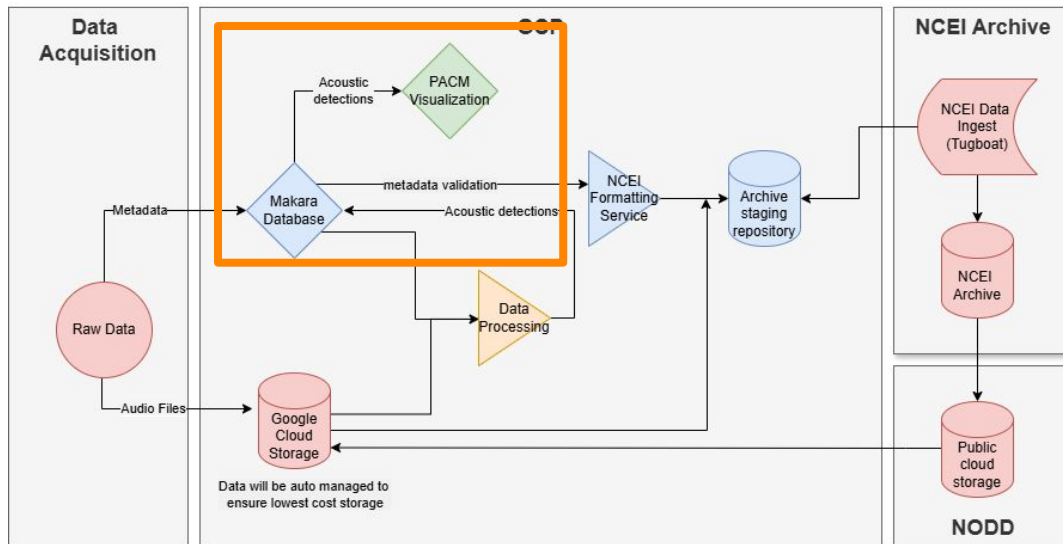
NEFSC PAB | Walker Environmental Research

Northeast Fisheries Science Center  
National Oceanic and Atmospheric Administration  
U.S. Department of Commerce



# Bigger Picture

**Goal:** create centralized database for **managing, sharing, and analyzing** PAM data in a consistent format and for supporting **timely data processing, research and management** across NOAA Fisheries



# Makara/PACM Data System

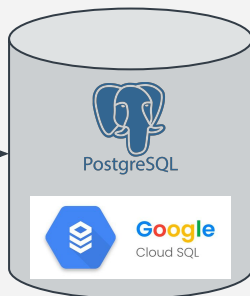
## Data Management

### Frontend Website

| ID  | Organization | Deployment Code     | Platform type           | Start      | End        |
|-----|--------------|---------------------|-------------------------|------------|------------|
| 744 | HEFRC        | MEFRC_SOW_200401_NH | SOTTOM_AJANTES_AJONGING | 2004-01-20 | 2004-01-16 |
| 743 | HEFRC        | MEFRC_SOW_200401_NH | SOTTOM_AJANTES_AJONGING | 2004-01-20 | 2004-01-16 |
| 742 | HEFRC        | MEFRC_SOW_200401_NH | SOTTOM_AJANTES_AJONGING | 2004-01-20 | 2004-01-16 |
| 740 | HEFRC        | MEFRC_SOW_200401_NH | SOTTOM_AJANTES_AJONGING | 2004-01-20 | 2004-01-16 |
| 739 | HEFRC        | MEFRC_SOW_200401_NH | SOTTOM_AJANTES_AJONGING | 2004-01-20 | 2004-01-16 |
| 693 | HEFRC        | MEFRC_SOW_200401_NH | SOTTOM_AJANTES_AJONGING | 2004-01-20 | 2005-03-14 |
| 740 | HEFRC        | MEFRC_SOW_200401_NH | SOTTOM_AJANTES_AJONGING | 2004-01-20 | 2005-03-14 |
| 747 | HEFRC        | MEFRC_SOW_200401_NH | SOTTOM_AJANTES_AJONGING | 2004-01-20 | 2005-03-14 |

Metadata, Detections,  
GPS/Sensor Data  
(CSV Templates)

## Data Storage



Makara



BigQuery Dataset

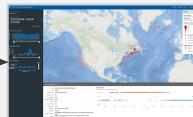
## Data Sharing and Release



NCEI Archive

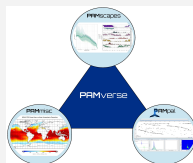


Data Requests



PACM

## Data Processing and Analysis



PAMverse

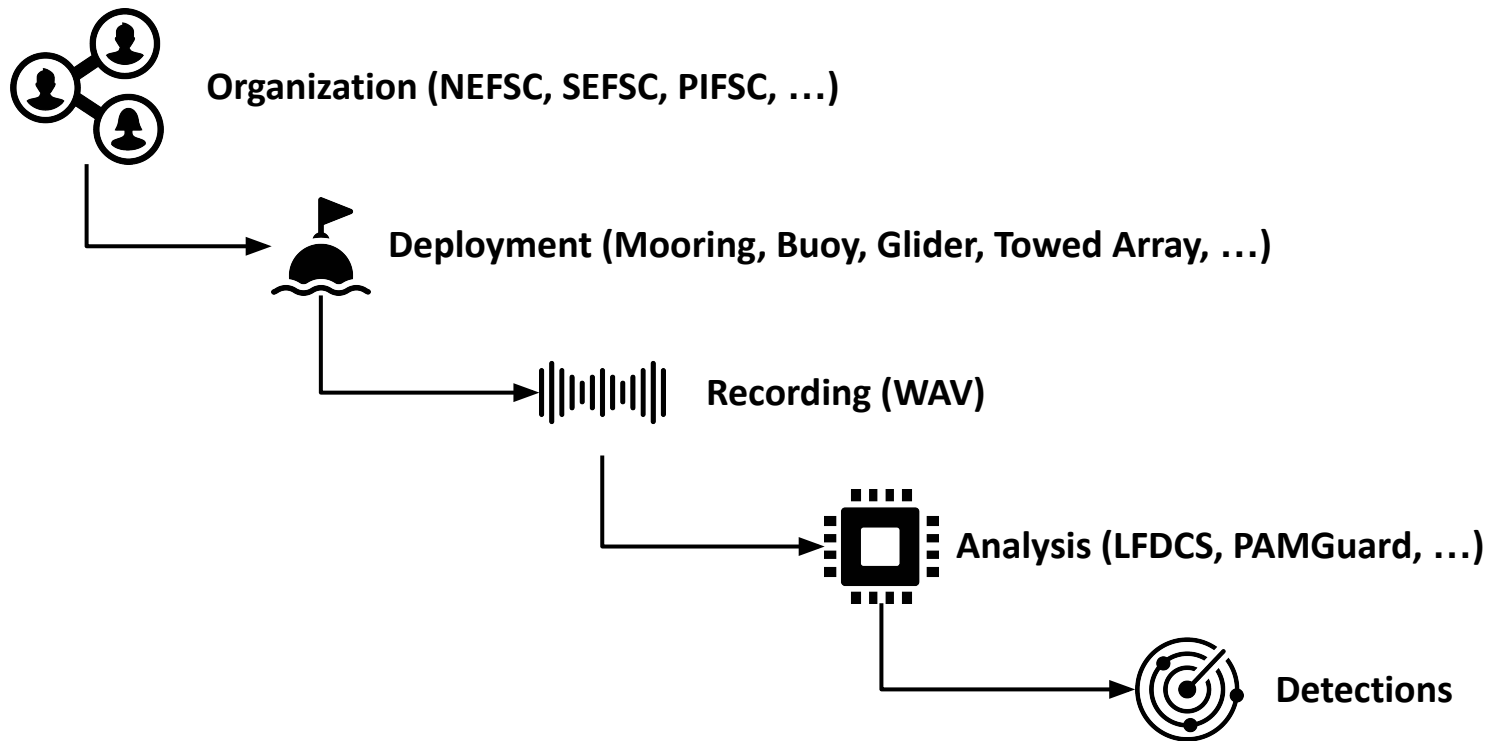


R / Python Scripts



Cloud Workstation  
(pam-ww)

# Core Database Schema



# Supplemental Datasets



Tracks (mobile platforms)



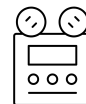
Sensor Measurements (e.g., temperature)



Recording



Hydrophone



Recorder



Projects (SANCTSOUND, NRS)



Sites (COX01, NRS09)



# Current Stats

**Deployments** 3,500

**Recordings** 3,824

**Analyses** 7,954

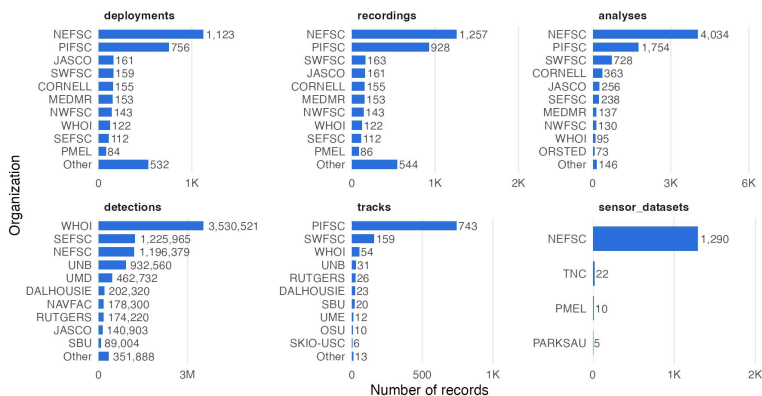
**Detections** 8,484,792

**Devices** 2,158

**Tracks** 1,097

**Sensor Datasets** 1,327

Makara records by table and organization (top 10)



Source: BigQuery ggn-nrms-pacm-dev-1.makara, 2026-09-24

## # Analyses per Species

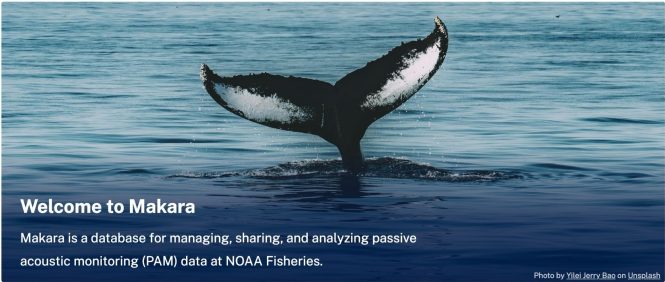
| Code  | Species                               | Analyses |
|-------|---------------------------------------|----------|
| HJWH  | Humpback Whale                        | 1,716    |
| FWH   | Fin Whale                             | 1,402    |
| RWH   | North Atlantic Right Whale            | 1,468    |
| SEWH  | Sei Whale                             | 1,248    |
| BLWH  | Blue Whale                            | 1,199    |
| SPWH  | Sperm Whale                           | 1,108    |
| UNDO  | Unid Dolphin                          | 944      |
| GOBW  | Goose-Beaked Whale                    | 754      |
| KWH   | Killer Whale                          | 679      |
| BLBW  | Blainville's Beaked Whale             | 661      |
| UNKO  | Unid Pygmy/Dwarf Sperm Whale          | 661      |
| PWDO  | Pacific White-Sided Dolphin           | 636      |
| UNBW  | Unid Beaked Whale                     | 627      |
| PSWH  | Pygmy Sperm Whale                     | 627      |
| DSWH  | Dwarf Sperm Whale                     | 627      |
| GRAM  | Risso's Dolphin                       | 615      |
| MWH   | Minke Whale (Common)                  | 598      |
| LBWH  | Longman's Beaked Whale                | 554      |
| PKW   | Pygmy Killer Whale                    | 549      |
| MEWH  | Melan-Headed Whale                    | 549      |
| FRDO  | Fraser's Dolphin                      | 549      |
| PSDO  | Pantropical Spotted Dolphin           | 549      |
| FKWH  | False Killer Whale                    | 549      |
| SPFW  | Pilot Whale (Short-Finned)            | 549      |
| DESP  | Unid Common Dolphin                   | 549      |
| SADO  | Short-Beaked Common Dolphin           | 549      |
| BODO  | Bottlenose Dolphin                    | 549      |
| RTDO  | Rough-Toothed Dolphin                 | 549      |
| STDO  | Striped Dolphin                       | 549      |
| CSDO  | Gray's Spinner Dolphin                | 476      |
| LBDO  | Long-Beaked Common Dolphin            | 476      |
| UNCE  | Unid Cetacean                         | 476      |
| UNLD  | Unid Large Dolphin                    | 476      |
| UNMD  | Unid Medium Delphinid                 | 476      |
| UNPD  | Unid Porpoise/Dolphin                 | 476      |
| NRWD  | Northern Right Whale Dolphin          | 476      |
| DAPO  | Dall's Porpoise                       | 476      |
| UNSC  | Unid Spinner/Clymene Dolphin          | 476      |
| BRWH  | Bryde's Whale (Includes Eden's Whale) | 380      |
| GRWH  | Gray Whale                            | 227      |
| PWH   | Unid Pilot Whale                      | 159      |
| BWC   | Cross Seamount Beaked Whale (BWC)     | 158      |
| UNME  | Unid Mesoplodon                       | 153      |
| KWHR  | Killer Whale (Transient)              | 130      |
| KWHR  | Killer Whale (Resident)               | 130      |
| KWHOF | Killer Whale (Offshore)               | 111      |
| GBW   | Gervais' Beaked Whale                 | 107      |
| HPO   | Harbor Porpoise                       | 103      |
| NBWH  | Northern Bottlenose Whale             | 86       |
| MMME  | Unid True's/Gervais' Beaked Whale     | 86       |
| SOBW  | Sowerby's Beaked Whale                | 86       |
| NBHF  | Narrow Band High Frequency Call       | 85       |
| HJRW  | Huber's Beaked Whale                  | 80       |
| BW43  | Unid Beaked Whale (BW43)              | 80       |
| BBWH  | Baird's Beaked Whale                  | 80       |
| STBW  | Stejneger's Beaked Whale              | 80       |
| GTBW  | Griego-Toothed Beaked Whale           | 78       |
| ASDO  | Atlantic Spotted Dolphin              | 73       |
| TRBW  | True's Beaked Whale                   | 73       |
| SRWH  | Southern Right Whale                  | 73       |
| SNDO  | Spinner Dolphin                       | 73       |
| RCWH  | Rice's Whale                          | 73       |
| OBDO  | Offshore Bottlenose Dolphin           | 73       |
| LPFW  | Pilot Whale (Long-Finned)             | 73       |
| CLDO  | Clymene Dolphin                       | 73       |
| BWG   | Gulf Of Mexico Beaked Whale (BWG)     | 73       |
| BOWH  | Bowhead Whale                         | 73       |
| BLU   | Beluga Whale                          | 73       |
| UNWH  | Unid Whale (Size Unknown)             | 73       |
| WBDO  | White-Beaked Dolphin                  | 73       |
| WSDO  | Atlantic White-Sided Dolphin          | 73       |

# Makara Frontend Website



**Makara Passive Acoustics Database**  
Signed in as jeffrey.walker@noaa.gov

Home Manage data ▾ Query database Documentation ▾ Contact Us Admin Dashboard



**Welcome to Makara**

Makara is a database for managing, sharing, and analyzing passive acoustic monitoring (PAM) data at NOAA Fisheries.

Photo by Yllei Jerry Bao on Unsplash

For more information about Makara and the related data ecosystem, see the [Lab Manual](#).

**Documentation and Reference Information**

**[Lab Manual](#)**  
A field guide to the NOAA Fisheries passive acoustics data ecosystem, including Makara, PACM, and PARS, and how they work together to support passive acoustics research and management.

**[Database Schema](#)**  
Learn about the tables, relationships, identifiers, and controlled vocabularies that make up the Makara data model.

**[Data Templates Guide](#)**  
The definitive reference for each CSV template, with every field, data type, and validation rule, plus blank templates and example submissions.

**[Reference Tables](#)**  
Browse the controlled vocabulary codes for species, sound sources, platforms, instruments, and other categorical fields.

- **Documentation**
  - Lab Manual
  - Data Templates Guide
- **Manage Data**
  - *Database Tables*: Create/Read/Update/Delete Individual Records in each Table
  - *Import Data*: CSV Templates -> DB
  - *Export Data*: DB -> CSV Templates
  - *NCEI Archive*: DB + GCS Audio Files -> NCEI
- **Query Database**
  - Filter metadata, detections, and other data by primary attributes (platform type, species, etc.) across all organizations
  - Joins tables to create simplified views of the data
  - Download as CSV files ready for analysis
- **Contact Us**
  - Request access or get help

<https://makara-462670382708.us-east4.run.app>

# Live Demo



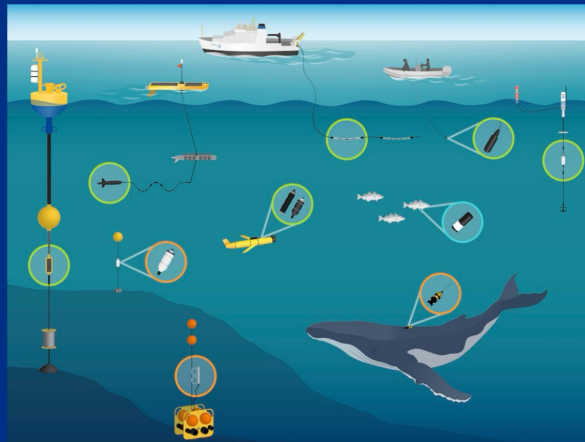
Passive Acoustics

Home PACM PARS Makara

## Listen: Do you hear that?

Passive acoustic technologies allow us to study the behavior and movements of marine animals, their contribution to the ocean soundscape, and how they are affected by human-made sounds.

This website provides tools for storing and visualizing passive acoustic monitoring (PAM) data of the ocean.

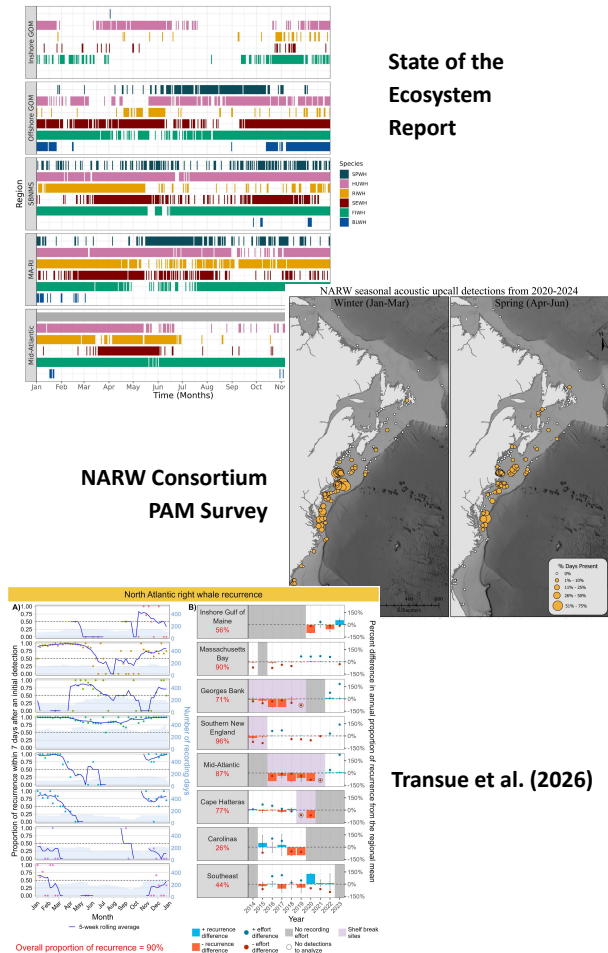


NOAA Fisheries studies marine animals by using a variety of technologies to record underwater sounds, including archival passive acoustic recordings (orange), real-time acoustic data collection (green), and active acoustics (blue).

<https://passiveacoustics.fisheries.noaa.gov>

# Efficiencies and Applications

- **State of the Ecosystem Report** - figure summarizing all NEFSC archival data for 6 species across all regions
- **NARW Consortium PAM Survey Summaries** - summary figures of all PAM activities across multiple organizations
- **Transue et al. (2026)** - developed “A standardized framework for quantifying species occurrence, applied to passive acoustic monitoring of baleen whales”
- **Zeh et al. (submitted)** - data integration for ecological data and long-term trends
- **Hansen et al. (in prep)** - looking at slow zones using all historical real time detections from Makara
- **Holdman et al. (in prep)** - summary figures of multiple species presence in GOM (pre-Makara, but more easily generated today)

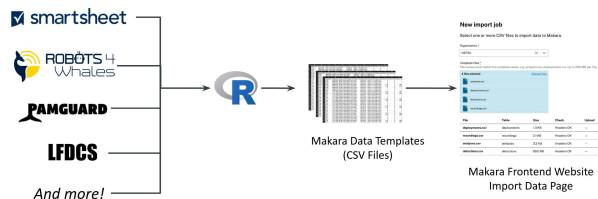


# Software Integration

- **PACM** - public data visualization tool displays deployment metadata and detection data (unless restricted by data owner)
- **PAMverse** (Taiki Sakai) - R packages to visualize PAM detection data reads Makara exports directly from BigQuery and/or CSV files
- **PyPAM** - auto-generate YAML attributes file
- **AI/ML Annotations Database (Dan Woodrich)** - Uses recording metadata and URIs to generate vector embeddings of audio files
- **PAM Workstations** - Access recording metadata via BigQuery directly from workstation to support analytical workflows

# Getting Started

- Request Makara user account via Contact Us form
  - Makara account linked to NOAA Google account, no password needed
- Review the Lab Manual and Data Templates Guide
- Format your data using Makara data templates
  - Need help building a pipeline to automatically transform data? Talk to Taiki!



- Import template files using Import Data page
  - Tip: use {makaraValidatr} R package to validate template files before upload
- Data Updates and Corrections
  - Use Database Tables pages to make quick updates (one row at a time)
  - Export entire table, revise multiple rows, re-import (batch update)

# Who Can Access My Data?

All deployment and recording metadata shown on PACM and accessible to all Makara users within NOAA through database queries

All analysis metadata (detector, species, etc) accessible to all Makara users within NOAA through database queries

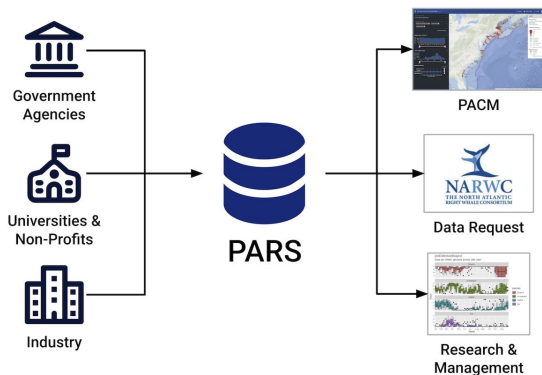
Detections can be restricted using release flags in analysis table:

- **analysis\_release\_data** - allow detections to be included in NARWC data requests
- **analysis\_release\_pacm** - allow detections to be shown on PACM
- **analysis\_release\_noaa** - allow detections to be accessible to other Makara users within NOAA through Query page or BigQuery

see ***Access & Permissions*** page in ***Lab Manual*** for more details

# PARS

Data submission system  
for **external collaborators**  
to contribute data to PACM

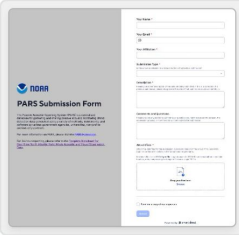


**NOAA  
FISHERIES**

Passive Acoustics

HomePACMPARSMakea

## Passive Acoustic Reporting System



### Looking to Submit Your Data?

First, please review the submission guide to learn about the data formats and other requirements. When you are ready to submit your data, use the submission form to upload your files.

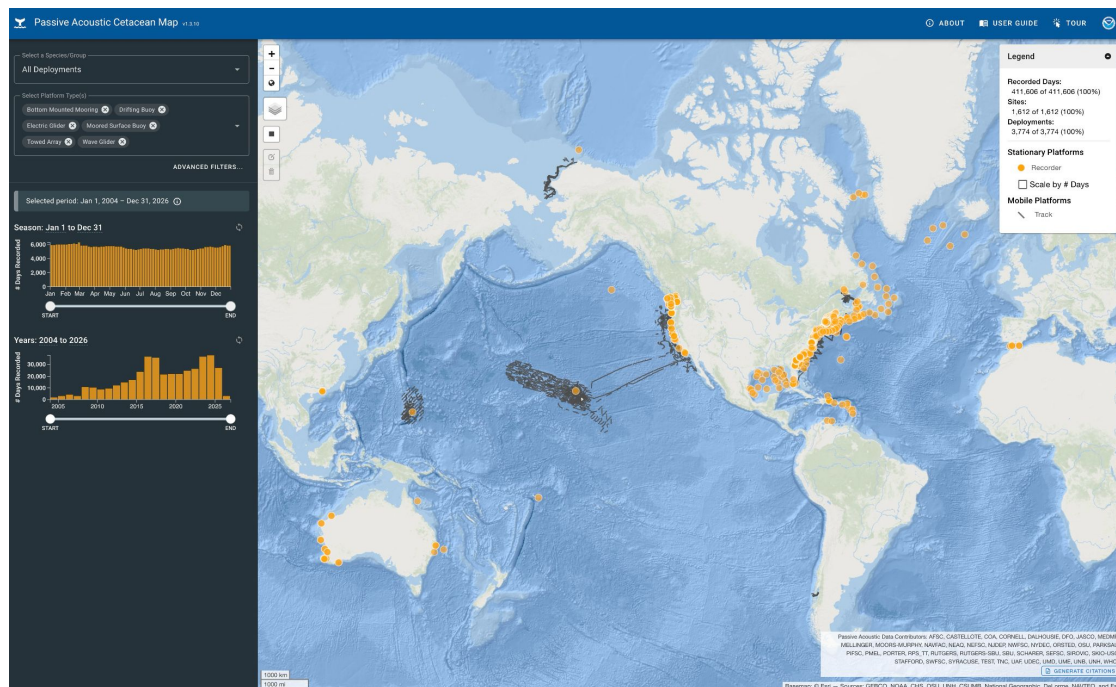
[View Submission Guide](#)[Open Submission Form](#)

The Passive Acoustic Reporting System (PARS) is a centralized database for gathering and sharing passive acoustic monitoring (PAM) detection data generated using a variety of methods, instruments, and software by various government agencies, universities, non-profits and industry partners.

<https://passiveacoustics.fisheries.noaa.gov/pars>

# PACM Updates

- Deployments grouped by site
- DMP filter
- Automated and user-provided citations
- Project funding



<https://passiveacoustics.fisheries.noaa.gov/pacm>

# Next Steps

- Request your accounts
  - Use Contact Us page on Makara Website
- Makara Workshop (mid-October)
  - Deep dive into how Makara works, how to use it, and how to format your data
- Makara Office Hours (TBA)
  - Drop in to get help or ask questions
- Currently In Development
  - NCEI Archiving Pipeline
  - Data automations
    - Data Ingest (e.g. real-time detections)
    - Transfer from Makara -> PACM
- We're here to help!

# Questions?

<https://passiveacoustics.fisheries.noaa.gov>

